

Intelligent Management. Seamless Connectivity.



InCloud Manager

· AI Network Assistant

· Zero-Touch Cloud Management

· Cloud-Native Management

· Remote Deployment

1. Product Overview

Intelligent Management. Seamless Connectivity.

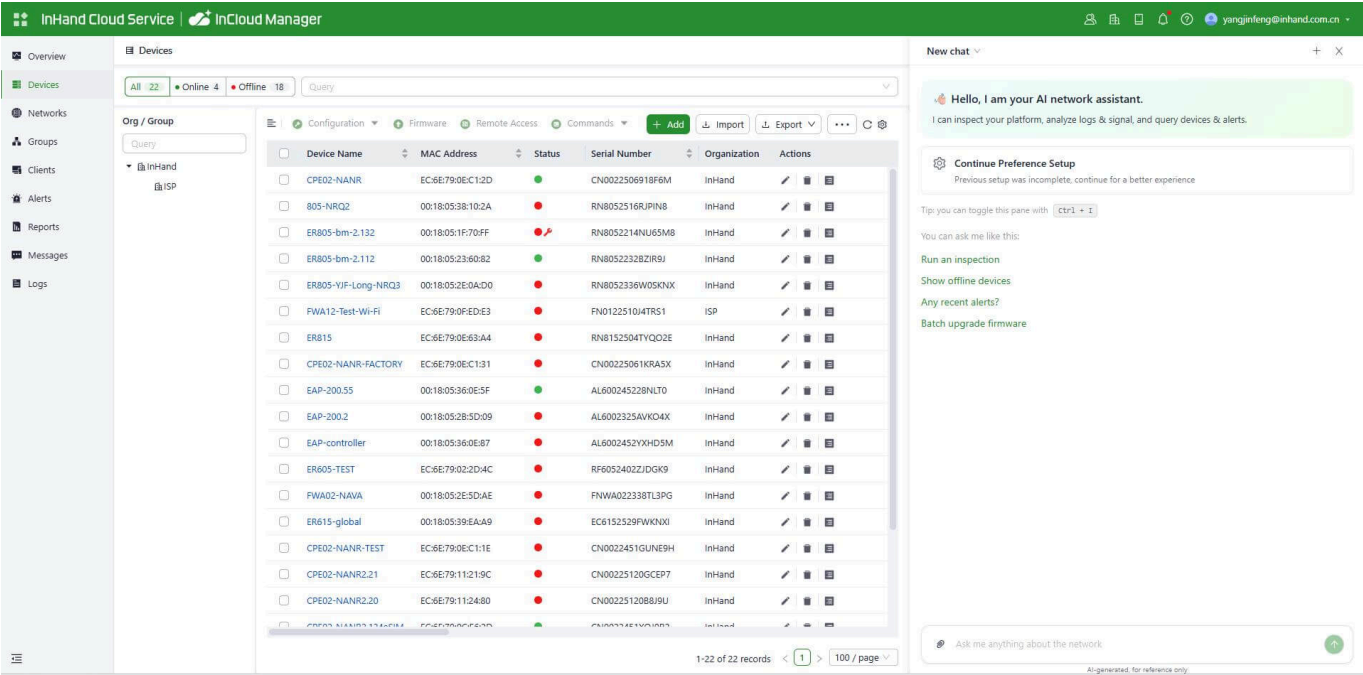
AI Network Assistant · Zero-Touch Cloud Management

InHand Networks InCloud Manager is an enterprise network management platform designed for retail stores and SMB offices. Combined with InHand edge routers, APs, and switches, it delivers a simplified, cloud-native network management solution. InCloud Manager enables businesses to move device and network operations to the cloud, supporting one-stop remote deployment and centralized management of network and communication devices across any scenario.

The newly launched **AI Network Assistant** brings network operations into the natural language era — issue plain-language instructions to complete device inspections, fault diagnosis, and batch configurations, dramatically improving operational efficiency. It helps businesses grow effectively, reduce management costs, and strengthen their competitive advantage in digital operations.

2. AI Network Assistant

Simply type a natural language instruction in the AI Network Assistant dialog to manage the full device lifecycle — no page-by-page navigation required, no professional network engineering background needed.



3. The O&M Challenge: Staffing Constraints and Efficiency Bottlenecks

Common pain points in traditional network management:

Channel Partners	End Customers
One person can only manage a limited number of devices — scaling up means falling behind	Routine inspection of dozens of devices requires hours of clicking through Web UIs
Experienced network engineers are scarce and costly	Troubleshooting an offline device means jumping through multiple pages — slow and tedious

Channel Partners	End Customers
Customers are geographically dispersed, making on-site visits expensive	Signal quality requires manual lookup and comparison against standards
	Batch operations are done entirely by hand, device by device

4. 9 AI Capabilities Covering the Full Device Lifecycle

Device Monitoring

Get real-time visibility into every device's operational status — no manual page browsing required.

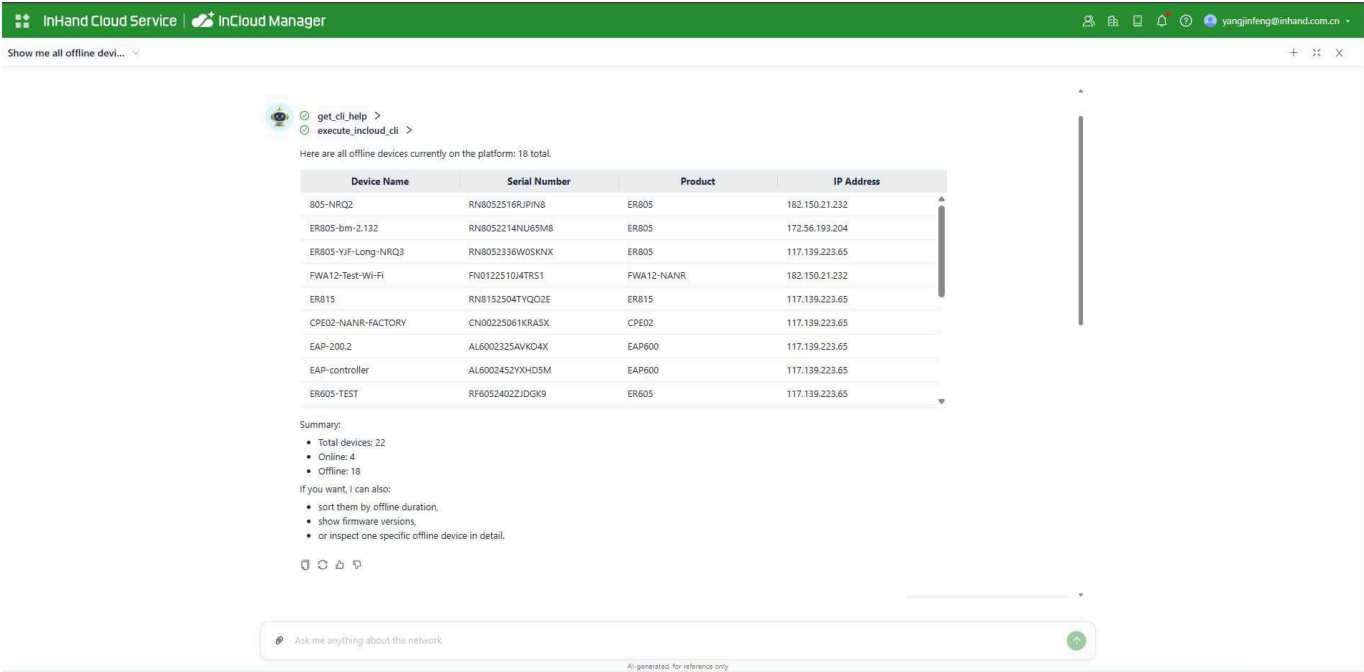
- Query online / offline status with filtering by name, location, or group
- View online history to track recent device stability trends
- Real-time traffic statistics to detect abnormal bandwidth consumption
- View connected client count and details

Example:

User: Show me all offline devices

Assistant:

Device Name	SN	Offline Since
Router A	SN001	5 minutes ago
Router B	SN002	5 minutes ago



Fault Diagnosis

More than a simple query — the AI automatically runs a multi-step diagnostic chain and delivers a complete analysis with actionable recommendations.

Example: **Device Offline Diagnosis** — AI executes 5 automated steps:

Step	AI Action	Finding
1. Check device status	Confirm offline timestamp	Last online at 10:23
2. Review online history	Analyze recent disconnect patterns	3 dropouts in the past week, mostly at night
3. Check signal records	Analyze RSRP trend	Dropped from -95 to -112 dBm (weak signal range)
4. Review system logs	Search for key events	Modem reset events found

Step	AI Action	Finding
5. Conclusion	—	"Persistent signal degradation causing frequent disconnects — recommend inspecting antenna"

Example:

User: Why did the first device go offline?

Assistant: Auto-checked signal → reviewed logs → Conclusion: Signal degradation (RSRP -112 dBm). Recommend checking the antenna.

Let's analyze the log to f...

Let's analyze the log to figure out what exactly caused this.



analyze_logs >

write_todos >

fetch_device_info_by_sn >

write_todos >

Got device info: wsc9 is currently online. Pulling cloud syslog for the last 24 hours and the 09:05 offline event window.

download_cloud_syslog >

Analyzing the cloud syslog window that covers the reported period, including the event near 2026-04-01 09:05 Asia/Shanghai.

task >

write_todos >

I analyzed the device logs for **wsc9**. The most likely cause is:

Conclusion

The **offline event around 09:05** was most likely caused by a **device reboot**, and the repeated short disconnects around it are consistent with an **unstable cellular uplink under weak radio conditions**.

Key evidence from logs

1) Reboot happened just before the offline period

- 2026-04-01 09:00:52 — device rebooted
- 2026-04-01 09:00:54 — wan cellular1 is not up!
- 2026-04-01 09:01:08 — interface cellular1, changed state to up
- 2026-04-01 09:01:08 — cellular(sdma) is connected

This strongly indicates the device went offline because it restarted, and then the cellular WAN came back shortly after boot.

2) Another short cellular outage happened right after

- 2026-04-01 09:08:53 — raw send error(113: Host is unreachable)

Ask me anything about the network.

AI-generated, for reference only.

Signal Analysis

No need to manually look up standards or compare raw values — the AI evaluates cellular signal quality and delivers a clear, direct verdict.

Signal Level	RSRP Range	Description
Excellent	> -85 dBm	Strong signal, stable operations
Good	-85 ~ -100 dBm	Normal operation
Fair	-100 ~ -110 dBm	Potential risk, recommend monitoring

Signal Level	RSRP Range	Description
Poor	< -110 dBm	High risk, immediate action recommended

- Automatically flags devices that fall below professional thresholds
- View historical signal trends to detect degradation before it causes issues
- Cross-device signal quality comparison to quickly identify underperforming sites

Example:

User: Which devices have poor signal quality?

Assistant: 3 devices currently have poor signal quality:

Device	RSRP	Level	Recommendation
Store A – Router	-118 dBm	Poor	Check antenna or adjust mounting position
Store C – Router	-113 dBm	Poor	Check antenna
Warehouse B – Router	-106 dBm	Fair	Continue monitoring

The screenshot shows the InHand Cloud Manager web interface. At the top, there's a green header with the InHand logo and navigation icons. Below the header, a chat window is open. The user's message is: "Which devices have poor signal quality?". The AI assistant's response is: "Based on the available cellular signal readings, these devices stand out as having poor signal quality: 1. CPE02-NANR-TEST: SN: CH00224510JNE3H, Last signal: -110 dBm, Level: 2/4, Network: 4G LTE, Last reported: 2025-09-22T10:19:55Z, Status: Offline. 2. CPE02-404: SN: CH002245117VRL6, Last signal: -101 dBm, Level: 3/4, Strength: 2, Network: 4G LTE, Last reported: 2025-07-16T03:30:00Z, Status: Offline. This is borderline poor / weak. Devices with weaker-than-ideal but not clearly poor: CPE02-NANR2.124eSIM — -90 dBm, ER805-bm-2.132 — -87 dBm, CPE02-NANR2.21 — -87 dBm. Notes: Many devices currently have no signal data in their latest state, so they can't be reliably classified from this list alone. Most poor/weak readings above are also stale because the devices are offline. If you want, I can test: list all cellular devices sorted by signal strength, or..."

Remote Diagnostics

Run diagnostic commands directly without logging into the device Web UI — results returned instantly.

- **Ping:** Test network connectivity and latency
- **Traceroute:** Trace path and locate breakpoints
- **Packet Capture:** Capture packets for protocol analysis
- **Speed Test:** Measure actual bandwidth and verify link quality

Example:

User: Run a speed test on Store A's router cellular interface

Assistant: Speed test complete. Results:

- Download: 48.3 Mbps
- Upload: 12.1 Mbps
- Latency: 34 ms / Packet Loss: 0%
- Conclusion: Network is operating normally — current link meets business requirements

The screenshot displays the InCloud Manager web interface. At the top, a green header bar contains the 'InHand Cloud Service' and 'InCloud Manager' logos, along with user information 'yangjinfeng@inhand.com.cn'. Below the header, a search bar shows the query 'run a network speed tes...'. The main content area features a chat interface with a bot icon on the left. The chat history includes three commands: 'execute_incloud_cli >', 'get_cli_help (3 times) >', and 'execute_incloud_cli (2 times) >'. The bot's response states: 'Done — I ran a speed test on ER805-bm-2.112 over its cellular interface.' It then lists the results: 'Interface: Cellular(SIM1) (rnet_apn1)', 'Band: B2', 'Server: WebNX Inc, New York, NY', 'ISP: Cisco IoT', 'Idle latency: 39.84 ms', 'Jitter: 7.24 ms', 'Download: 31.19 Mbps', 'Upload: 17.16 Mbps', and 'Packet loss: 0.0%'. It also shows data usage: 'Download phase: 56.50 MB' and 'Upload phase: 30.70 MB'. A footer note says 'If you want, I can also check the device's historical speed test results or compare this with signal quality.' At the bottom, there is a text input field with the placeholder 'Ask me anything about the network' and a green send button. A small disclaimer 'AI-generated, for reference only.' is visible at the very bottom.

Configuration Management

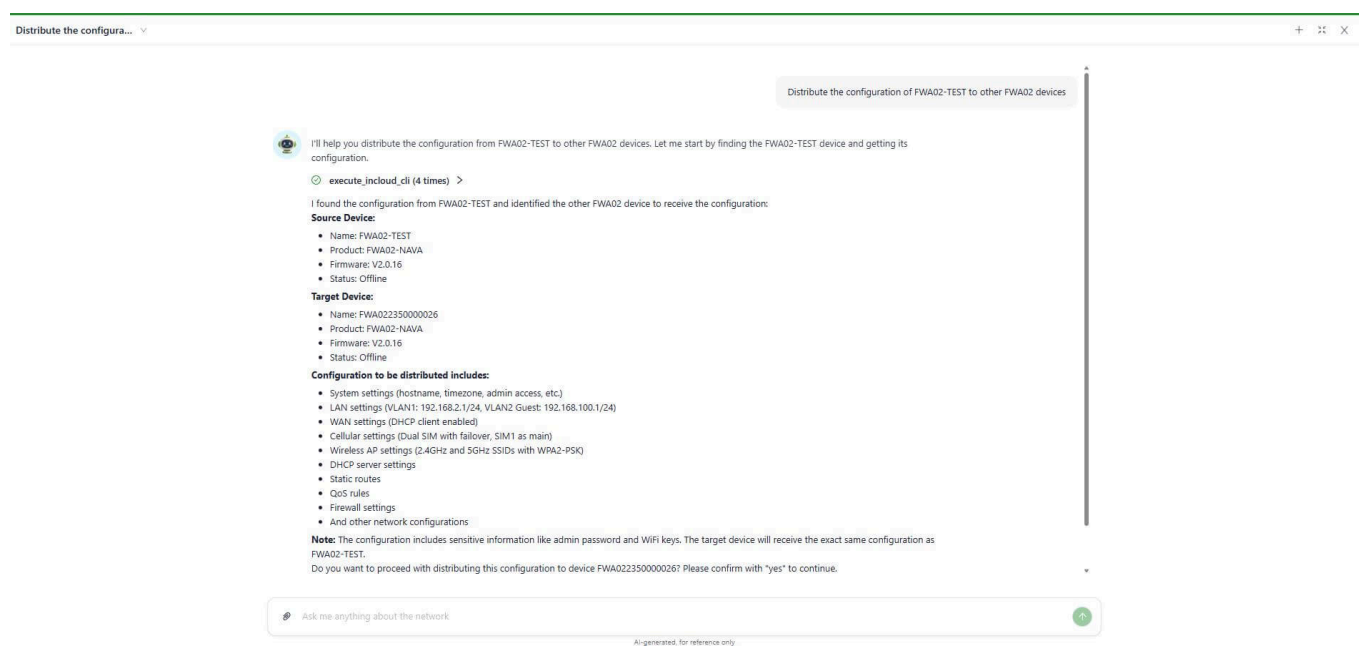
Read, modify, and copy device configurations with full traceability and rollback support at every step.

- **Read / Modify:** Query or update device configuration items directly, without opening the Web UI
- **Copy Config:** Quickly replicate one device's configuration to other devices
- **Snapshot Backup:** Automatically save a configuration snapshot before any change
- **Rollback:** Restore to any previous configuration version in one click

Example:

User: Use Device-xxx's config as the standard template and push it to all other ER805 devices

Assistant: Current config comparison → Diff review → Confirm changes → Awaiting confirmation before execution



Firmware Upgrade

AI manages the entire process from version selection to upgrade completion, significantly reducing the risk of manual errors.

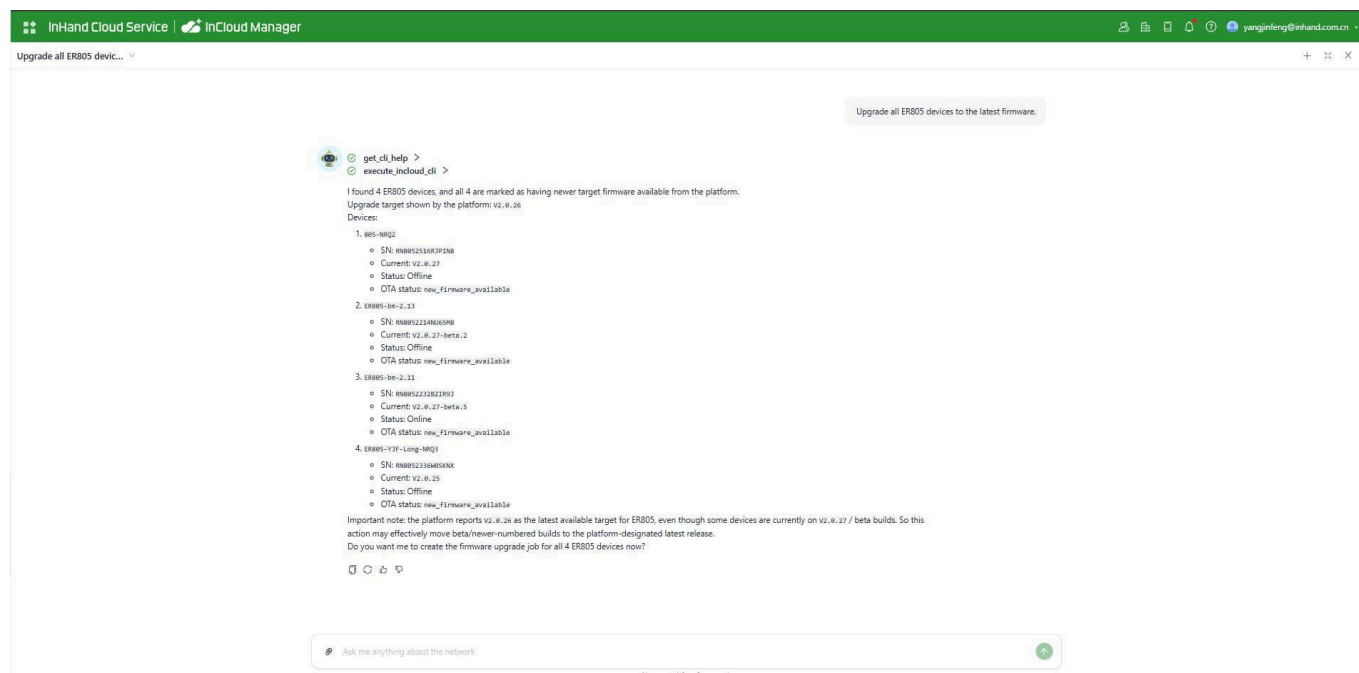
- **Compatibility Check:** Automatically verify firmware compatibility with device model before upgrading

- **Staged Rollout:** Support group-based or time-window-based batch upgrades to minimize business risk
- **Progress Monitoring:** Track upgrade status for every device in real time with automatic alerts on failure
- **Rollback Protection:** Instantly revert to the previous firmware version if an upgrade fails

Example:

User: Upgrade all ER805 devices to the latest firmware

Assistant: Detected 23 ER805 devices. Current version: V2.0.18. Latest version: V2.0.27. Compatibility check: All passed. Recommended: 3 batches of ~8 devices each, with 10-minute intervals. Estimated total time: 35 minutes. Confirm to proceed?



Alert Management

Centralized view and management of all network alerts — no anomaly goes unnoticed.

- View current unacknowledged alert list, sorted by severity
- Acknowledge / close alerts with action logging
- Manage alert rules: add, modify, enable or disable alert policies
- Correlated analysis of alerts and device status

Example:

User: What unacknowledged alerts are there recently?

Assistant: 5 unacknowledged alerts currently:

Device	Alert Type	Time	Severity
Store G – Router	Device Offline	Today 09:15	Critical
Warehouse B – Router	Cellular Data Overrun	Yesterday 23:40	Warning
Store D – Router	High CPU Utilization	Yesterday 18:22	Warning
Store B – Router	Link Status Abnormal	Yesterday 14:05	Info
Store F – Router	Signal Quality Degraded	Yesterday 11:30	Info
Would you like to handle them one by one, or prioritize a specific device?			

What are the unresolved...

What are the unresolved alerts recently?

Based on the current Alerts page, there are **42 unresolved (unconfirmed) alerts**.
Most recent visible unresolved alerts:

1. **2026-04-03 09:25:52 — Client Offline**
Device: **RL8052121000060**
Description: Client **7F:A0:5D:21:67:79** has been offline for more than 10 minutes.
2. **2026-04-02 17:48:44 — Client Offline**
Device: **RL8052121000060**
Description: Client **0A:51:0E:23:9C:19** has been offline for more than 10 minutes.
3. **2026-04-02 17:45:34 — Client Offline**
Device: **RL8052121000060**
Description: Client **3E:3C:D2:8E:7E:94** has been offline for more than 10 minutes.
4. **2026-04-02 17:43:39 — Client Offline**
Device: **RL8052121000060**
Description: Client **A7:09:C9:1A:BE:27** has been offline for more than 10 minutes.
5. **2026-03-31 18:07:16 — Client Offline**
Device: **RL8052121000060**
Description: Client **27:97:A9:2B:F7:C2** has been offline for more than 10 minutes.

Other unresolved alert types visible on the page include:

- **Wired WAN Connected**
- **Poor Cellular Signal Strength**
- **High Memory Utilization**
- **High Average CPU Utilization**

Quick takeaway:

- The **latest unresolved alerts** are mainly "Client Offline"
- They are concentrated on device **RL8052121000060**

If you want, I can also help you:

- group the unresolved alerts by **type**
- list only alerts from the **last 7 days**
- identify which alerts look **most critical** first

Ask me anything about the network

AI-generated, for reference only

Remote Terminal

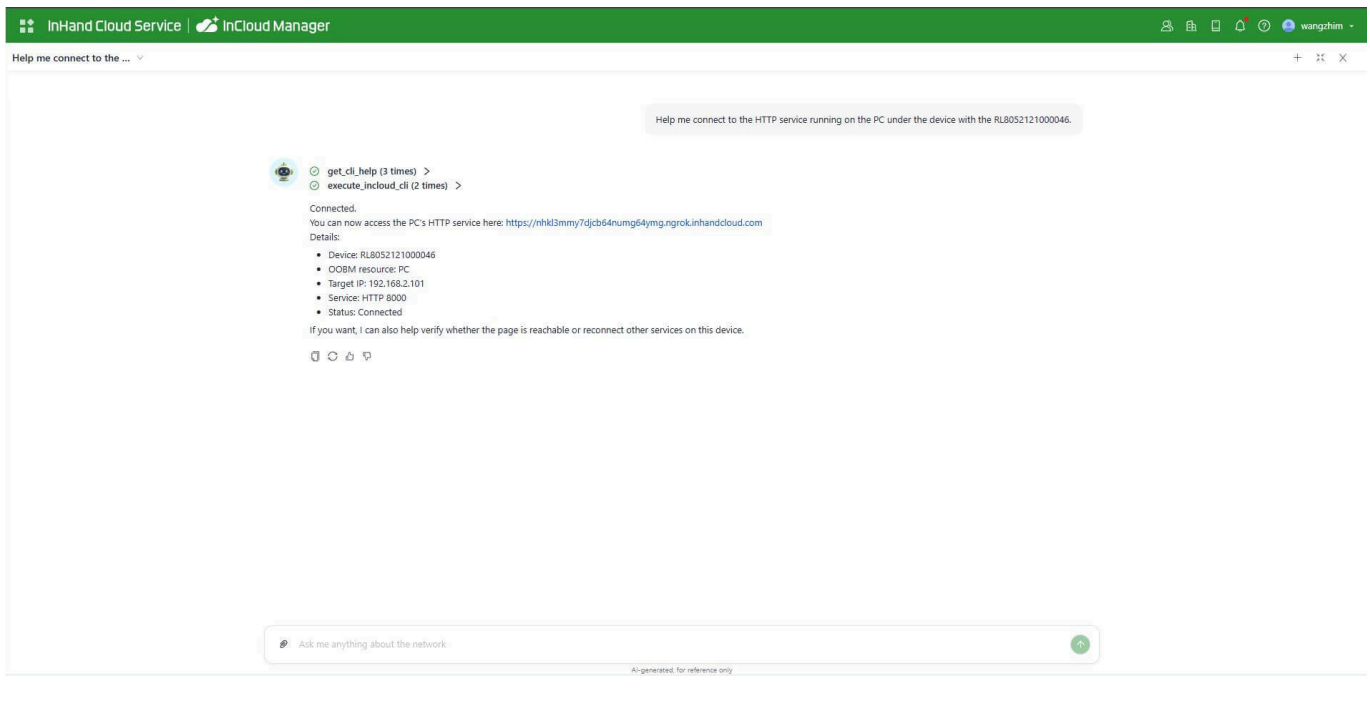
For advanced operations, launch the device control interface directly — no manual navigation required.

- **Open Device Web UI:** Jump directly to the target device's management page
- **Open Command Line:** Access the device CLI directly for advanced operations
- All operations are authenticated by user account; activity logs are fully auditable

Example:

User: Open the management interface for Store A's router

Assistant: Remote connection established. Store A (SN: ER805-001) Web UI is now open in a new window. This session has been recorded in the activity log.



Batch Operations

Issue instructions to multiple devices simultaneously — no repetitive manual device-by-device execution.

- **Batch Inspection:** Run a health check on all devices at once and generate a consolidated report
- **Batch Config Push:** Push a configuration template to a specified device group
- **Batch Firmware Upgrade:** Initiate bulk firmware upgrades with unified progress tracking
- **Batch License Management:** View license status, renew, or assign licenses in bulk

Example:

User: Run a monthly inspection on all retail store devices

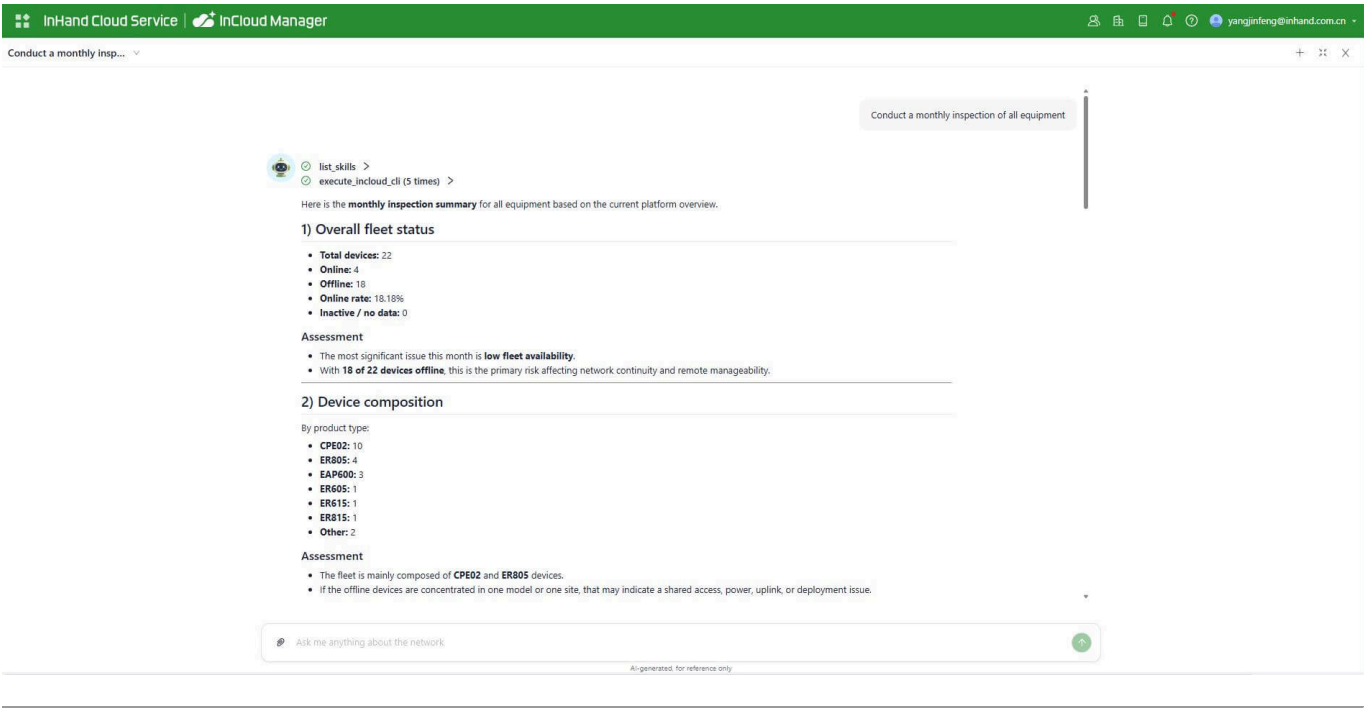
Assistant: Inspecting 60 devices... Complete. Report:

- Online and healthy: 42 devices
- Weak signal (attention needed): 3 devices (Store A, C, F)
- Offline: 18 device

...

- I can continue with one of these:
 1. Inspect the 18 offline devices
 2. Check which device has config error / pending config
 3. Review firmware candidates for upgrade
 4. Focus on a specific device from the list

If you want, I can start by drilling into the **offline devices** first.



5. AI-Powered O&M: Across-the-Board Efficiency Gains

Task	Traditional Approach	With AI Network Assistant
Monthly device inspection	Log into each device's Web UI individually — takes days	Natural language instruction — complete report in minutes

Task	Traditional Approach	With AI Network Assistant
Offline device troubleshooting	Browse multiple pages manually, make your own judgment	AI auto-runs full diagnostic flow, delivers complete conclusion
Batch config deployment	Manual filtering, device-by-device execution	Issue one instruction, batch execution runs automatically
Bulk firmware upgrade	Manual device-by-device operation, manual progress tracking	Unified initiation, staged execution, automatic progress reporting
Signal quality audit	Manually look up data and compare against standards	AI auto-evaluates signal quality and delivers direct conclusions

Reduce Labor Costs: Team members don't need deep network expertise — AI effectively closes the experience gap. New staff get up to speed quickly, significantly shortening training time.

Reduce Travel Costs: Most issues can be identified and resolved remotely. On-site visits are reserved for scenarios where physical presence is truly necessary.

Generate Additional Revenue (for MSPs / System Integrators): Inspection reports can be delivered directly as client service deliverables, forming a standardized, billable value-added service.

6. Transparent and Controllable — Every Step Visible

Every AI operation is fully transparent. Not a black box — a visible, accountable assistant.

Safety Mechanism	Description
✓ Read-only operations unrestricted	Query status, view alerts, analyze signal — zero risk, use anytime
✓ Write operations require confirmation	Modify config, upgrade firmware, reboot — changes displayed in full before execution, awaiting user confirmation
✓ Operations tied to user account	Every action is logged with the operator's identity; complete activity audit trail
✓ Standardized execution flow	Each operation follows the same process every time, regardless of who performs it — reducing human error risk

7. Industry Applications

Applicable wherever distributed devices need centralized management.

Industry	Typical Scenario	Core Value
Retail Chain	Convenience stores, cafes, pharmacies	Network downtime directly impacts sales — fast recovery minimizes revenue loss
ATM / Financial	Bank branches, payment terminals	Strict compliance requirements and SLA commitments
Industrial / Energy	Production lines, power stations, mining sites	Remote O&M reduces downtime and ensures production continuity
Traffic Monitoring	Highways, urban traffic systems	Centralized batch management of hundreds or thousands of devices

Industry	Typical Scenario	Core Value
MSP / System Integrator	Multi-customer managed services	Manage multiple clients with one person; inspection reports become billable services

Case Study: Retail Chain Store Network Operations *(Each store's router handles POS + surveillance + digital price tags + HQ connectivity)*

Scenario	Traditional Approach	With AI Network Assistant
Store outage — manager reports network down	Remote page-by-page investigation, time-consuming	AI locates root cause in minutes → Significantly reduces business downtime
Monthly inspection — days of device-by-device review	Manual device-by-device audit	AI generates complete report in minutes → Problem stores automatically flagged
Pre-event network check — ahead of major promotion	Manual monitoring, ad hoc	Real-time monitoring during event period → O&M team has full visibility and confidence
New store rollout — push standard config to all new devices	Manual device-by-device configuration	Natural language batch instruction → Dramatically reduces labor investment